



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: DATA ANALYTICS AND VISUALIZATION – (24DSB3101)

A.Y. 2026 - 2027

TITLE OF THE PROJECT	Explainable Real Estate Analytics: A Comparative Analysis of Machine Learning Frameworks for House Price Prediction	
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Abstract

The real estate industry generates large volumes of property-related data that can be leveraged to estimate house prices accurately. Traditional valuation methods often rely on expert knowledge and manual analysis, which may produce inconsistent results. This project proposes a machine learning-based framework for House Price Prediction by performing comprehensive Exploratory Data Analysis (EDA), data preprocessing, feature engineering, and comparative evaluation of multiple machine learning models.

Initially, the collected housing datasets undergo preprocessing, including handling missing values, removing duplicate records, encoding categorical attributes, and normalizing numerical features. A detailed Exploratory Data Analysis (EDA) is then conducted to understand the data through dataset overview, descriptive statistics, data quality assessment, missing value analysis, duplicate detection, outlier detection, univariate analysis, bivariate analysis, multivariate analysis, correlation analysis, feature engineering, and feature selection.

Multiple machine learning algorithms, including Decision Tree, Random Forest, XGBoost, and CatBoost, are trained and evaluated for predicting house prices. Their performance is compared using MAE, MSE, RMSE, R² Score, and MAPE, and the best-performing model is selected.

To improve model interpretability, the project incorporates SHAP and LIME, providing both global and local explanations of predictions. These techniques help identify the most influential features affecting house prices.

Finally, the analytical results are presented through interactive visualizations such as heatmaps, boxplots, feature importance graphs, correlation plots, and dashboards. A comparative analysis is conducted across all machine learning models to determine the most accurate and interpretable approach for house price prediction.

Signature of the Guide

HOD Signature